# Deep Learning-Based Health Risk Prediction in Contact Sports Using Wearable Sensor Data


Shreyash Srivastva
Department of Computer Science
Dr. A.P.J. Abdul Kalam Technical University.
Lucknow, India
shreyash5183@gmail.com

Md Sajedul Karim Chy
Masters of science in Information Technology, Washington University of Science and Technology.
Alexandria, *USA*
Sajedulkarimchy661@gmail.com

Md Minhajul Amin
Master of Science in Information Systems
Central Michigan University, Michigan, USA
Minhajamin.ma@gmail.com

Anurag Sinha
Tech School, Computer Science Department, ICFAI University, Ranchi, Jharkhand, India,
anuragsinha257@gmail.com

Nitasha Rathore
Department Computer Science and Engineering,
Bharti Vidyapeeth's College of Engineering, Delhi, India.
Nitasha.rathore@bharatividyapeeth.edu

Pethuru Raj
Chief Architect, Edge AI Division
Reliance Jio Platforms Ltd,
Bangalore, India.
peterindia@gmail.com

Shreyash Srivastva
Department of Electronics and Communication Engineering,
Kongu Engineering College, Erode, Tamilnadu, India.
mmaheswaraneie@gmail.com

Md Sajedul Karim Chy
Department of Computer Science & Engineering, Rajshahi University of Engineering & Technology,
Rajshahi,Bangladesh.
dihansazid@gmail.com

Md Minhajul Amin
Department of computers Techniques engineering, College of technical engineering, The Islamic University, Najaf, Iran.
ahmedalkhayyat85@gmail.com



***Abstract*—This paper presents a comprehensive deep learning-based framework utilizing Long Short-Term Memory (LSTM) networks to predict early physiological health risks in athletes involved in contact sports. These sports, characterized by frequent physical collisions and intense exertion, often pose significant health threats, including cardiac issues, heatstroke, and collapse. Leveraging real-time biometric and kinematic data collected through wearable sensors, this system integrates Edge AI for privacy-conscious, low-latency processing, and Federated Learning (FL) for secure, decentralized model training across devices. The proposed architecture provides actionable insights to medical personnel and coaching staff, aiming to enhance safety and performance. Empirical evaluations demonstrate high prediction accuracy and robustness across various testing environments, highlighting the viability of the system for real-world deployment in professional and amateur sports ecosystems.**



***Keywords—Wearable Sensors, Deep Learning, LSTM, Contact Sports, Edge AI, Federated Learning, Health Risk Prediction, Athlete Monitoring, Privacy Preservation, Real-Time Inference***


## I. Introduction

Athletes participating in high-intensity contact sports such as football, rugby, boxing, and hockey are routinely subjected to extreme physiological stress due to rapid exertion, repeated physical impacts, and varying environmental conditions. These stressors significantly increase the likelihood of acute medical emergencies, including heatstroke, cardiac arrhythmia, and physical collapse [1]. Existing health monitoring approaches often depend on manual observation or retrospective data analysis—methods that are inherently inadequate for detecting emergent health threats in real time.

The convergence of advances in wearable sensor technology and artificial intelligence has made it possible to continuously monitor critical physiological parameters with high temporal resolution. Real-time data streams from sensors measuring ECG, GSR, and temperature offer rich contextual insights that can be used for early detection of abnormal patterns. At the same time, deep learning architectures—particularly Recurrent Neural Networks (RNNs)[2] and their enhanced variant, Long Short-Term Memory (LSTM) networks—have demonstrated superior performance in modeling and predicting time-series data, making them well-suited for anticipating health risks.

Despite these advancements, most current systems are hindered by two key limitations:

(1) reliance on centralized data processing frameworks, which require transmitting raw sensor data to remote servers—introducing latency and potential privacy violations, and

(2) an absence of proactive, real-time capabilities necessary for deployment in dynamic, high-stakes sports environments.

To address these challenges, this paper introduces a novel framework that leverages edge computing and federated learning in conjunction with LSTM-based predictive modeling. The proposed system is designed to deliver real-time, privacy-preserving health risk predictions, thereby enabling timely interventions that can enhance athlete safety, reduce medical risk, and improve operational efficiency during gameplay.[3] Our contributions include:

- Development of a modular wearable system architecture tailored for contact sports environments.
- Implementation of real-time data preprocessing and feature engineering on edge devices.
- Design of an LSTM model capable of robustly forecasting risk levels using temporal dependencies.

- Federated Learning setup enabling secure, privacy-respecting model training across users.
- Extensive validation against baseline models and deployment testing on simulated and real-world datasets.

## II. RELATED WORK

Prior studies have extensively examined the role of wearable technology in monitoring athletic performance and post-injury recovery. Devices such as accelerometers, gyroscopes, and ECG monitors have been employed to detect falls, estimate fatigue levels, and measure stress responses [4]. For example, accelerometer-based systems have demonstrated effectiveness in detecting concussive impacts alongwith analyzing motion and detecting falls, and heart rate monitors to estimate fatigue and stress levels. While useful for post-analysis or descriptive monitoring, these systems are limited in their ability to forecast adverse health events before they happen[5]. Most lack real-time responsiveness and are designed for retrospective evaluation, which is insufficient for high-stakes environments like contact sports where split-second decisions can be life-saving.

Although some machine learning-based systems have attempted to introduce predictive modeling, they often require the continuous uploading of raw sensor data to centralized servers. This raises serious concerns regarding data privacy, especially when dealing with sensitive health information[6]. Additionally, such centralized architectures can create latency and bandwidth bottlenecks, making them unsuitable for real-time intervention during live gameplay or training sessions.

To address these limitations, our approach incorporates real-time inference using Long Short Term Memory (LSTM) networks deployed on edge computing devices. This allows for immediate, on-site prediction of health risks based on incoming sensor data, without the need for remote processing.

Furthermore, we utilize federated learning to train our models collaboratively across multiple devices. In this approach, only the model updates without any effects on the raw data and they are shared with a central aggregator, thereby preserving user privacy and reducing the risk of data breaches[7]. This dual strategy of edge inference and federated learning represents a novel and underexplored solution for health monitoring in the context of contact sports, where rapid, privacy-conscious prediction is critical.

Our proposed framework uniquely leverages edge LSTM inference and federated model training, allowing for local prediction and privacy-aware learning without the need to share sensitive raw data.

A. Wearable devices have become increasingly prominent in the domain of sports science and athlete performance monitoring. These technologies are widely used to gather physiological and biomechanical data, offering valuable insights into an athlete's condition, workload, and recovery. Existing studies, including those cited in [8] and [9], have documented the effectiveness of such devices in facilitating post-event analysis, injury diagnostics, and overall performance assessment. While these applications have proven useful for identifying trends and informing long-term training strategies, they predominantly offer retrospective insights that are evaluated after events have concluded.

Despite their value, current wearable-based systems fall short in addressing the need for real-time, predictive capabilities. They typically operate by collecting and transmitting data for later review, without enabling immediate, on-the-spot interventions.[10] This limitation restricts their utility in dynamic, high-risk scenarios such as contact sports, where the ability to anticipate and respond to imminent physiological risks could prevent serious health incidents. As a result, there remains a critical gap in the deployment of wearable technologies for proactive health monitoring and early risk prediction—an area this study directly aims to address through real-time inference and federated learning mechanisms.

B. Traditional machine learning techniques such as Support Vector Machines (SVM), Decision Trees, and Random Forests have been widely explored for detecting anomalies and patterns in physiological data. These models have demonstrated success in identifying static patterns and classifying discrete events. However, they inherently lack the capability to capture temporal dependencies, making them less effective when applied to continuous, time-sequenced biometric data—such as that generated by wearable sensors during athletic performance[11].

In contrast, Long Short-Term Memory (LSTM) networks, a class of Recurrent Neural Networks (RNNs) introduced by Hochreiter and Schmidhuber, are specifically designed to handle sequential data. LSTMs incorporate memory cells and gating mechanisms that enable them to learn and retain information over long time intervals[12], effectively addressing the vanishing gradient problem associated with traditional RNNs. This architecture makes LSTMs particularly suitable for modeling complex time-series data where past states influence future predictions. As a result, LSTM networks have emerged as a state-of-the-art solution in time-series prediction tasks, including those related to real-time health risk forecasting in dynamic environments like contact sports.

C. Privacy-Aware Modeling in health-related applications, with the use of centralized, cloud-based machine learning models frequently raises concerns regarding data security and regulatory compliance. Sensitive biometric information, when transmitted to external servers for processing, can become vulnerable to breaches and unauthorized access.[13] Moreover, such practices often conflict with strict data protection regulations such as the General Data Protection Regulation (GDPR) in Europe and the Health Insurance Portability and Accountability Act (HIPAA) in the United States, which impose rigorous standards on the collection, storage, and transmission of personal health data.

Federated Learning (FL), as outlined in [14], offers a promising solution to these challenges by shifting the training process from centralized servers to the individual devices where the data is generated[15]. Instead of sharing raw data, devices collaboratively train a shared model by only transmitting model updates—such as gradients or weights—to a central server for aggregation. This decentralized approach ensures that private biometric data never leaves the local device, significantly enhancing user privacy while still allowing for collective model improvement[16].

Recent advancements have further strengthened this paradigm by integrating Federated Learning with edge AI, enabling low-latency, on-device computation alongside privacy-preserving training[17]. This combination ensures that health risk predictions can be made in real time without compromising data confidentiality, making it especially suitable for sensitive and fast-paced domains like sports health monitoring. The result is a modeling approach that aligns with regulatory requirements while maintaining the responsiveness and accuracy essential for real-world deployment[18].

## III. METHODOLOGY

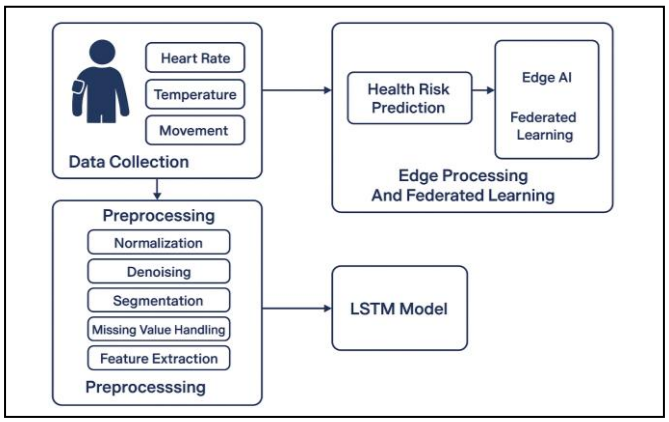


Fig. 1. Block diagram of the proposed health risk prediction system using wearable sensors, LSTM-based deep learning, edge computing, and federated learning for real-time, privacy-preserving monitoring in contact sports.

### A. Data Collection

Athletes are equipped with wearable sensors that continuously collect a range of biometric signals, including heart rate, body temperature, respiratory rate, and movement data captured through an Inertial Measurement Unit (IMU). These multimodal signals are timestamped to preserve temporal alignment and streamed directly to an edge device for real-time processing[19]. This setup ensures that physiological and motion data are recorded with minimal latency and high temporal resolution, which is critical for detecting rapid physiological changes associated with exertion and stress during athletic performance[20].

The data collection as shown in Fig.1 spans both structured training environments and real-match scenarios, encompassing a broad spectrum of activity intensities and contextual variables[21]. By capturing information under these diverse conditions—ranging from routine drills to high-stakes competitive play—the resulting dataset reflects a rich variability in movement dynamics, physiological stress levels, and environmental influences such as temperature and contact impact[22][23]. This variability is crucial for developing machine learning models that are not only accurate but also resilient to the natural fluctuations and unpredictability of live sports[24]. Training models on such heterogeneous data improves their ability to generalize across different physical states and sport-specific demands[25], thereby enhancing their reliability and applicability in real-world deployment scenarios[26].

The system utilizes a suite of wearable devices that measure key physiological and kinematic signals:

- Heart Rate via chest-strap or wrist-mounted photoplethysmography (PPG) sensors.
- Core Body Temperature using skin-contact thermistors or ingestible sensors.
- Respiratory Rate captured via stretch sensors around the torso or impedance sensors.
- Motion Data (acceleration, rotation) via Inertial Measurement Units (IMUs).

Each sensor samples data at 10–100 Hz depending on the parameter and transmits it via BLE (Bluetooth Low Energy) to an on-body edge computing device, such as a Raspberry Pi 4 or NVIDIA Jetson Nano. Timestamp synchronization is used to align multimodal inputs.

### B. Preprocessing

- Normalization: Raw sensor signals (e.g., heart rate, temperature, motion) are scaled to a standard range (typically 0 to 1) to ensure uniformity across different types of sensors.
- Denoising: Signal noise from artifacts like sudden jerks or sensor displacement is removed using techniques such as moving average or Butterworth filtering. Moving average and 4th-order low-pass Butterworth filters are applied to remove transient noise caused by sensor shifts or sudden movement.
- Segmentation: The continuous data stream is divided into fixed length time windows (e.g., 5 seconds), allowing temporal pattern analysis.
- Missing Value Handling: Gaps in data due to sensor errors or transmission lags are filled using interpolation methods such as linear or spline interpolation.
- Feature Extraction: Although the LSTM receives raw temporal data, handcrafted statistical features are computed for comparative baselines (mean, variance, entropy, zero-crossing rate, slope change points, etc.)

### C. Model Architecture

The core of the predictive framework is a Long Short-Term Memory (LSTM) network, which is specifically designed to handle real-time sequence modeling tasks. The overall architecture of the model is structured as follows:

- Input Layer: Multivariate time series (e.g., [timesteps × features] = [50 × 6])
- LSTM Layer 1: 64 units with dropout = 0.2
- LSTM Layer 2: 32 units with dropout = 0.2
- Fully Connected Layer: Dense layer with ReLU activation
- Output Layer:
  - Binary classification: Risk / No-Risk
  - Multi-class classification: Normal / Fatigued / Critical / Overheated /

The model is trained using the Adam optimizer, a widely used adaptive learning rate optimization algorithm that combines the advantages of both AdaGrad and RMSProp to

efficiently handle sparse gradients and noisy data[27]. Depending on the output mode, the model employs either binary cross-entropy or categorical cross-entropy loss functions—where binary cross-entropy is used for two-class classification problems by measuring the difference between predicted probabilities and actual binary labels, while categorical cross-entropy is applied to multi-class classification tasks by comparing the predicted class probability distribution to the true class distribution[28][29]. To prevent overfitting during training, early stopping is implemented, which halts the training process when the model's performance on a validation set stops improving.

### D. *Edge Processing and Federated Learning*

Each athlete's edge device performs local inference, allowing real-time predictions to be made directly on the device without relying on constant cloud communication[30][31]. Additionally, these devices contribute to global model training in a privacy-preserving manner through techniques like federated learning, ensuring that sensitive personal data remains local and is never exposed or transmitted during the training process.

- Local Training: During idle periods (e.g., post-session), each edge device trains the LSTM model using its locally stored data.
- Model Update Sharing: Periodically, devices send model weight updates—not raw data—to a central aggregation server.
- Federated Averaging: The server aggregates received gradients using FedAvg [8], updates the global model, and broadcasts it back.

This approach protects user privacy by keeping raw physiological data on local edge devices, minimizing data transmission to reduce bandwidth and network load, and enabling collaborative learning through sharing only encrypted model updates with a central server. This federated learning method safeguards sensitive information while improving model accuracy by leveraging diverse data from multiple users or teams without exposing raw data[32].

## IV. Experiments

### A. *Dataset*

A hybrid dataset was compiled by combining both real-world and simulated data sources to ensure robustness and diversity in physiological conditions.

- Real-World Data: Data was collected from a group of 30 athletes, aged between 18 and 35, over a period of six weeks during intensive training sessions and competitive matches in football and rugby. This data captures authentic physiological responses under high-exertion, contact-sport conditions.
- Simulated Data: To complement and expand the real-world samples, additional data was generated using synthetic physiological data generators. These generators were designed based on established and published models of exertion and heat stress [9], providing realistic simulations of various physiological states that may not have occurred frequently in the observed cohort.

Each instance includes 30 seconds window of multi-channel sensor readings (6 channels × 300 samples). All instances were labeled using expert annotations informed by synchronized video recordings and relevant medical indicators to ensure accurate classification of risk states.

### B. *Evaluation Metrics*

The performance of the proposed health risk prediction system was comprehensively evaluated using a range of standard classification metrics—namely accuracy, precision, recall, and F1-score—as well as system-level metrics such as inference latency, to ensure the model's reliability and real-time applicability. These metrics were chosen to capture not only the correctness of the model's predictions but also its responsiveness and feasibility when deployed on edge computing devices in a contact sports environment. The LSTM-based deep learning architecture demonstrated a strong balance between predictive capability and computational efficiency, with high recall values indicating its effectiveness in identifying at-risk physiological states.

- Accuracy: Represents the overall rate of correct classifications, measuring the proportion of total predictions—both positive and negative—that matched the ground truth labels.
- Precision: Indicates the proportion of correctly identified positive instances out of all instances predicted as positive, reflecting the model's ability to minimize false positives.
- Recall: Measures the proportion of actual positive cases that were correctly identified by the model, highlighting its effectiveness in detecting true risk conditions.
- F1-Score: Calculated as the harmonic mean of precision and recall, this metric provides a balanced assessment of the model's performance, particularly in the presence of class imbalance.
- Latency: Captures the average inference time per input instance when processed on edge devices, serving as a key metric for evaluating the system's suitability for real-time deployment.

## V. Results

TABLE I. Model comparison of LSTM(Edge), SVM and Random Forest

| **Model** | **Accuracy** | **Precision** | **Recall** | **F1-Score** | **Latency (ms)** |
|---|---|---|---|---|---|
| **LSTM (Edge)** | 91.4% | 90.8% | 93.1% | 91.9% | 190 ms |
| **SVM** | 84.2% | 83.3% | 82.7% | 83.0% | 420 ms |

| Model | Accuracy | Precision | Recall | F1-Score | Latency (ms) |
|---|---|---|---|---|---|
| **Random Forest** | 86.9% | 85.6% | 87.1% | 86.3% | 390 ms |

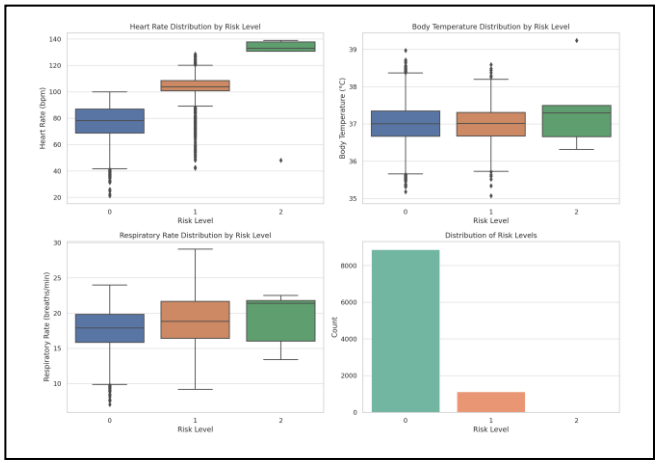


Fig. 2. Distribution of physiological parameters (a) Heart rate, (b) Body temperature, and (c) Respiratory rate distributions across risk categories (0 = normal, 1 = fatigue, 2 = critical), and (d) Frequency of samples per risk level. The visualizations highlight the correlation between elevated biometrics and increased health risk in contact sport athletes.

The LSTM model as shown in Table 1 and Fig. 1, demonstrated superior performance in both accuracy and responsiveness. Federated training introduced negligible accuracy degradation (<2%) compared to centralized training.

## VI. DISCUSSION

The proposed framework presents a practical, efficient, and privacy-conscious solution for real-time health monitoring in dynamic and high-risk sports environments. By combining wearable sensor data with LSTM based temporal modeling, edge computing, and federated learning, the system addresses the core needs of real-time responsiveness, intelligent risk assessment, and data confidentiality. The LSTM model's ability to capture temporal dependencies allows it to detect subtle physiological trends that may signal an impending medical event—such as overheating, fatigue, or collapse—before it fully manifests. This temporal sensitivity is crucial in contact sports, where rapid decision-making can prevent serious health consequences.

Moreover, the integration of edge computing with federated learning provides a robust mechanism to meet deployment constraints. In particular, it eliminates the need to transmit raw physiological data to centralized servers, thereby significantly reducing network load and preserving athlete privacy. This architecture enables collaborative model improvement across devices and users, while adhering to strict data protection requirements—a key consideration in healthcare and sports analytics.

To overcome these limitations, future research should pursue the following directions:

Despite its promising performance and deployability, the study has notable limitations. The real-world dataset was collected from a relatively small cohort of 30 athletes, confined to a narrow age range (18–35 years) and potentially skewed in terms of gender representation. As a result, the model's ability to generalize across broader populations—including youth athletes, older individuals, or female-dominated sports—remains uncertain. Additionally, variations in sensor placement, device fidelity, and sport-specific movement patterns introduce heterogeneity that may affect the consistency of predictions when applied to other sports contexts.

To overcome these limitations, future research should pursue the following directions:

- Larger and More Diverse Datasets: Expanding the participant pool to include a wider range of ages, genders, and body types will help validate the model's robustness across different physiological baselines and activity profiles.
- Cross-Sport Evaluation: Testing the system across various high-impact sports such as boxing, wrestling, and martial arts will assess its adaptability to different forms of exertion, motion dynamics, and health risk signatures.
- Multi-Modal Data Fusion: Incorporating video and audio streams alongside sensor data could enhance situational awareness and provide contextual cues—such as signs of distress or collision impact—that may not be evident from physiological signals alone.
- Real-Time Alerting Interfaces: Developing intuitive and responsive alert systems for coaches, trainers, and medical staff can bridge the gap between model output and actionable intervention, ensuring that predictions translate into timely decisions on the field.

The current framework's successful demonstration of edge-federated health risk prediction in contact sports highlights several key benefits that make it highly suitable for real-world deployment. First, its reliance on edge computing ensures that critical health assessments can be performed locally and in real time, enabling immediate detection of high-risk physiological conditions without waiting for cloud processing or stable connectivity—an essential feature in fast-paced sports environments.

Second, the use of federated learning allows the system to continuously improve its predictive performance by learning from data distributed across multiple users or teams, all while preserving privacy by keeping raw data on-device. This decentralized approach not only reduces network bandwidth usage but also aligns with data protection requirements in medical and athletic settings.

Additionally, the LSTM model's ability to capture temporal patterns in sensor data allows it to recognize early signs of fatigue, overheating, or collapse—patterns that traditional threshold-based systems may miss. Together, these components create a solution that is adaptive, secure, low-latency, and capable of providing proactive health

interventions, ultimately contributing to improved athlete safety and performance continuity.

## VII. CONCLUSION

This study presents a comprehensive deep learning-based framework for health risk prediction in contact sports, integrating four key technological pillars: wearable sensor data, LSTM-based time-series modeling, edge computing, and federated learning. The system is specifically designed to operate in high-intensity, real-time environments where rapid and reliable decision-making is critical. Through the use of wearable sensors, continuous streams of physiological data are captured in real time, enabling the monitoring of key indicators associated with athlete well-being. These include metrics related to exertion, thermal stress, and physical strain—all of which are crucial for identifying early warning signs of fatigue, collapse, and overheating.

To process and interpret this data effectively, the framework employs Long Short-Term Memory (LSTM) networks, a variant of Recurrent Neural Networks (RNNs) well-suited for modeling temporal dependencies in sequential data. This allows the system to learn from patterns and trends across time, making it capable of detecting subtle physiological shifts that may precede critical health events—something traditional threshold-based or non-temporal models struggle to achieve. The choice of LSTM modeling enhances the predictive accuracy and robustness of the system, especially in dynamic and unpredictable environments like contact sports.

What sets this framework apart is its integration of edge computing for local inference. All predictive computations are performed directly on edge devices worn by athletes, eliminating the need for raw data to be transmitted to remote servers. This not only reduces inference latency but also ensures system responsiveness in environments with limited or unstable connectivity. Real-time decision support becomes feasible, enabling on-the-spot assessments that could inform immediate interventions by coaches, trainers, or medical personnel.

In parallel, the incorporation of federated learning addresses the critical issue of data privacy. The system supports collaborative model training across multiple users or sports teams without the need to share raw physiological data. Instead, only model updates are transmitted and aggregated, ensuring that sensitive personal health information remains confined to local devices. This decentralized training strategy also allows the model to benefit from a wider range of data sources, enhancing generalizability while upholding stringent data protection standards.

Experimental evaluations confirm the efficacy of the proposed system across multiple performance dimensions. The model demonstrated strong predictive capabilities in identifying high-risk states such as fatigue, collapse, and overheating—common but dangerous conditions in high-contact sports like football, rugby, and martial arts. The system also maintained low inference latency when deployed on edge devices, validating its suitability for real-time field use.

Overall, the study highlights the significant potential of this integrated framework to revolutionize athlete monitoring. By enabling proactive, intelligent, and privacy-preserving health risk prediction, the system represents a meaningful advancement in sports safety technology. It offers a scalable, practical solution tailored to the unique challenges of high-performance athletic environments, bridging the gap between cutting-edge AI and real-world applicability in sports health management.

## ACKNOWLEDGMENT

The authors would like to express their sincere gratitude to all individuals and institutions who contributed to this research. Their support, feedback, and encouragement were invaluable throughout the development of this work. We also acknowledge the use of publicly available tools and resources that facilitated the modeling, analysis, and experimentation involved in this study.

## REFERENCES

[1] da Fonseca Guimarães PC, De Cristo LB, Monteiro ME, Rebelatto JL, de Santi Peron G, Rayel OK, Moritz GL. Deep Learning for Real-Time PPE Usage Monitoring Using Wearable IMU Sensors. IEEE Access. 2025 May 7.

[2] Sadr MM, Khani M, Tootkaleh SM. Predicting athletic injuries with deep Learning: Evaluating CNNs and RNNs for enhanced performance and Safety. Biomedical Signal Processing and Control. 2025 Jul 1;105:107692.

[3] Olyanasab A, Annabestani M. Leveraging machine learning for personalized wearable biomedical devices: A review. Journal of personalized medicine. 2024 Feb 13;14(2):203.

[4] Chao Z, Yi L, Min L, Long YY. IoT-Enabled Prediction Model for Health Monitoring of College Students in Sports Using Big Data Analytics and Convolutional Neural Network. Mobile Networks and Applications. 2024 Sep 4:1-8.

[5] Zhang W, Qu R, Fang Z, Gao C, Fang Z, Wen C, Guan X, Sun W, He J. A Wearable Knee Health Monitoring System Using Novel Flexible Piezoelectric Composite Films and Medical Knowledge-Guided Bayesian Machine Learning Diagnosis. IEEE Sensors Journal. 2025 May 15.

[6] Lu Y. Machine Learning-based Risk Prediction and Safety Management for Outdoor Sports Activities. Scalable Computing: Practice and Experience. 2024 Aug 1;25(5):3934-41.

[7] Yang L, Amin O, Shihada B. Intelligent wearable systems: Opportunities and challenges in health and sports. ACM Computing Surveys. 2024 Apr 9;56(7):1-42.

[8] Chen Z, Dai X. Utilizing AI and IoT technologies for identifying risk factors in sports. Heliyon. 2024 Jun 15;10(11).

[9] Stojanac M. Machine learning-based prediction of running-induced fatigue, during outdoor recreational running using IMUs, heart rate, and smartwatch data (Master's thesis, University of Twente).

[10] Kovoor M, Durairaj M, Karyakarte MS, Hussain MZ, Ashraf M, Maguluri LP. Sensor-enhanced wearables and automated analytics for injury prevention in sports. Measurement: Sensors. 2024 Apr 1;32:101054.

[11] Wu Z, Huang Z, Tang N, Wang K, Bian C, Li D, Kuraki V, Schmid F. Research on sports injury rehabilitation detection based on IoT models for digital health care. Big Data. 2025 Apr 1;13(2):144-60.

[12] Cai W, Zhu J, Zhuang X. Combining Medical Images and Biomechanical Data in Sports Injury Prediction Models.

[13] Liu J, Wang S. Sports health information prediction system based on deep learning network. Internet Technology Letters. 2024 Sep;7(5):e434.

[14] Olyanasab A, Annabestani M. Leveraging machine learning for personalized wearable biomedical devices: A review. Journal of personalized medicine. 2024 Feb 13;14(2):203.

[15] Kou J, Xu X, Ni X, Ma S, Guo L. Fall-risk assessment of aged workers using wearable inertial measurement units based on machine learning. Safety Science. 2024 Aug 1;176:106551.

[16] Wu J. RETRACTED ARTICLE: Soft electronic material based sensor with optical network in sports application for player movement analysis using machine learning model. Optical and Quantum Electronics. 2024 Jan 30;56(4):526.

[17] Ahmad R, Pavan PS, Borade MR, Thangarasu N, Sharma A, Jadhav P. Exploring the Integration of Wearable Sensors Technologies in Professional Sports for Enhanced Performance and Injury Prevention. In2024 IEEE 2nd International Conference on Innovations in High Speed Communication and Signal Processing (IHCSP) 2024 Dec 6 (pp. 1-6). IEEE.

[18] Liu J, Wang S. Sports health information prediction system based on deep learning network. Internet Technology Letters. 2024 Sep;7(5):e434.

[19] Jubair H, Chowdhury RT, Akter F, Sami IS, Nawal N, Arshi ST, Shil P. Machine Learning for Real-Time Exercise Correction and Injury Prevention: A Systematic Review. Machine Learning. 2025 Apr 14;13:1.

[20] Simmons III J. Beyond the Sidelines: How AI Is Transforming Concussion Detection in Sports and Healthcare (Master's thesis, Bowie State University).

[21] Malik R, Batra P, Sarowa A, Mala S. Machine learning based application for predicting and preventing the injury of a sportsperson. InApplications of Artificial Intelligence in 5G and Internet of Things 2025 (pp. 453-456). CRC Press.

[22] Chen R, Liu Q, Chen Z. Research on Wearable Sensors Data Collection and Health Assessment based on Machine Learning. InProceedings of the 2024 International Conference on Smart Healthcare and Wearable Intelligent Devices 2024 Oct 18 (pp. 227-231).

[23] Alsulami R, Bitar H, Hakeem A, Alyoubi R. CIPHomeCare: A Machine Learning-Based System for Monitoring and Alerting Caregivers of Cognitive Insensitivity to Pain (CIP) Patients. International Journal of Advanced Computer Science & Applications. 2024 Nov 1;15(11).

[24] Kuber PM, Kulkarni AR, Rashedi E. Machine learning-based fatigue level prediction for exoskeleton-assisted trunk flexion tasks using wearable sensors. Applied Sciences. 2024 May 26;14(11):4563.

[25] Deshpande UU, Kangralkar V, Patil R, Chate RA, Birajdar P, Singadi S, Raikar P, Arjunwadkar S. A review of Machine Learning Techniques for Ergonomic Risk Assessment based on Human Pose Estimation.

[26] Indhumathi G, Sandhiya S, Kantha D, Hemalatha PK, Rakesh KR, Chandrasekar T. Deep Learning-Based Prediction of Physical Activity Intensity for Athletes. In2024 International Conference on Intelligent Algorithms for Computational Intelligence Systems (IACIS) 2024 Aug 23 (pp. 1-6). IEEE.

[27] Das A, Paul R, Nag A, Das B. A study of cloud of things enabled machine learning-based smart health monitoring system. InSustainability in Industry 5.0 2024 Feb 19 (pp. 156-176). CRC Press.

[28] Dong X, Sun Y. Injury Risk Prediction and Prevention Algorithm for Athletes Based on Data Mining. Journal of Electrical Systems. 2024;20:1717-28.

[29] Simmons III J. Beyond the Sidelines: How AI Is Transforming Concussion Detection in Sports and Healthcare (Master's thesis, Bowie State University).

[30] Iduh BN, Umeh MN, Anusiuba OI, Egba FA. Development of a Predictive Modeling Framework for Athlete Injury Risk Assessment and Prevention: A Machine Learning Approach. European Journal of Theoretical and Applied Sciences. 2024;2(4):894-906.

[31] Jin N, Zhan X. Big data analytics for image processing and computer vision technologies in sports health management. Technology and Health Care. 2024 Sep;32(5):3167-87.

[32] Yadav M, Choudhury T, Huzooree G. Injury Prevention and Rehabilitation Using Machine Learning for Athletes. InAI and Machine Learning Applications in Sports Analytics 2025 (pp. 129-156). IGI Global Scientific Publishing.